4.3 Reflection Symmetry

Standing in front of a mirror, Kumaran was getting ready to celebrate his birthday. He noticed a beautiful sentence **I LOVE MOM** written on his T-shirt which was presented by his uncle.

In these words, he saw **I** and **MOM** were looking the same in the mirror. But the word **LOVE** did not appear the same.It looked as **ƎVOJ**.

Out of curiosity, he took out some alphabet cards and started checking which of the alphabets would look the same in the mirror. He found a few alphabets **A, H and I** look the same in the mirror, because they have lines of symmetry.

Already we know that a line of symmetry divides the figure into two equal halves. When you keep a mirror along the line of symmetry the other half of the figure gets reflected by the mirror and it looks the same. This is known as ***reflection symmetry*** or ***mirror symmetry***.



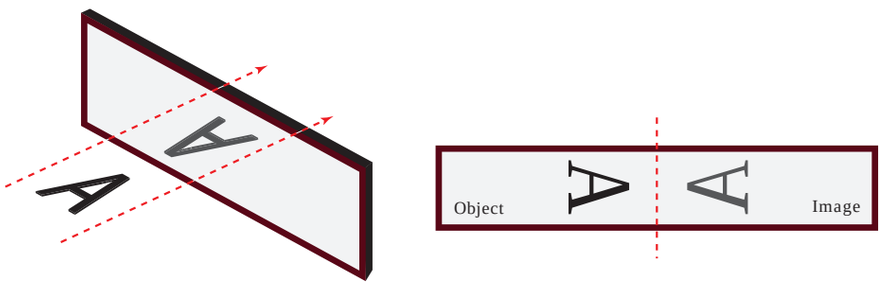
Think

Which other capital letters of English alphabets look like the same in the mirror?

Note

A shape has **reflection symmetry** if it has a **line of symmetry**.

When an object is seen in a mirror, the image obtained on the other side of the mirror is called its ***reflection***. The following figure shows the reflection of the English alphabet A. Let us assume that there is a line between A and its image in the place of a mirror.



E8MBAJ

We observe that an object and its mirror image are symmetrical with reference to the mirror line. If the paper is folded, the mirror line becomes the line of symmetry.

Note

- The object and its image will lie at the same distance from the mirror.
- The only difference is that the left side is on the right and vice-versa.

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